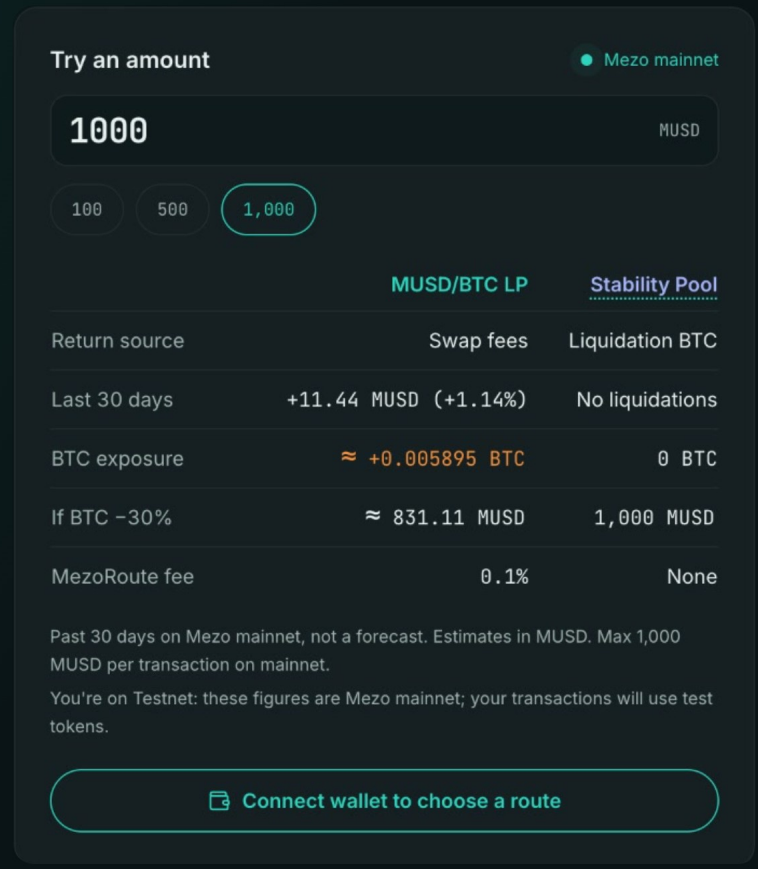
MEZOROUTE · BUILD WITH MUSD AND MEZO · WAVE 2

Pick a route. Know the BTC risk.

MezoRoute shows Mezo borrowers where to put their MUSD by what it does to their BTC risk, runs it in one transaction, and now manages the Trove behind it: unwind, close and leverage, plus MEZO staking.

Live on Mezo mainnet (capped) · every contract verified



mezoroute.vercel.app · github.com/hms1499/MezoRoute

SHIPPED IN WAVE 2

From a router to a Trove toolkit

Unwind, Close, Leverage

One transaction on your own Trove through a flash swap. The redemption queue is shown before and after.

MEZO staking, same transaction

The executor puts the new LP in the pool’s gauge for you. Claim, stake and unstake from Positions.

Enter from BTC

BTC in, no MUSD needed first. Same 10 bps fee, a gas reserve kept back, capped per transaction.

Plan before you connect

The LP planner works without a wallet. One design system, light and dark, every flow with a “Before you sign” panel.

Mainnet, verified: MezoRouteExecutorV2 0xb3461B...257915 · MezoRouteLeverage 0xA82059...6C6

THE PROBLEM

Borrowed MUSD has two homes and no map

01

Which route?

The MUSD/BTC pool and the Stability Pool earn in different ways and fail in different ways. Nothing puts them side by side.

02

Too many steps

Borrow, swap and add liquidity are separate screens, approvals and transactions. The swap size is a guess.

03

Hidden BTC risk

Borrowing against BTC, then holding half the LP in BTC, stacks BTC exposure. A BTC drop hits the Trove and the LP at once.

MezoRoute answers all three on one screen: compare, preview, execute.

Persona: a Mezo borrower who holds, or can borrow, MUSD against BTC

MEASURED IN THE MEZO APP

Borrow → LP today: 6 wallet confirmations

Screen	Wallet confirmation	Count
Borrow	Borrow more MUSD	1
Swap	Approve MUSD, swap to BTC	2
Pool	Approve MUSD, approve BTC, deposit	3
Total	3 screens	6

MEZOROUTE BORROW & DEPLOY

1 tx

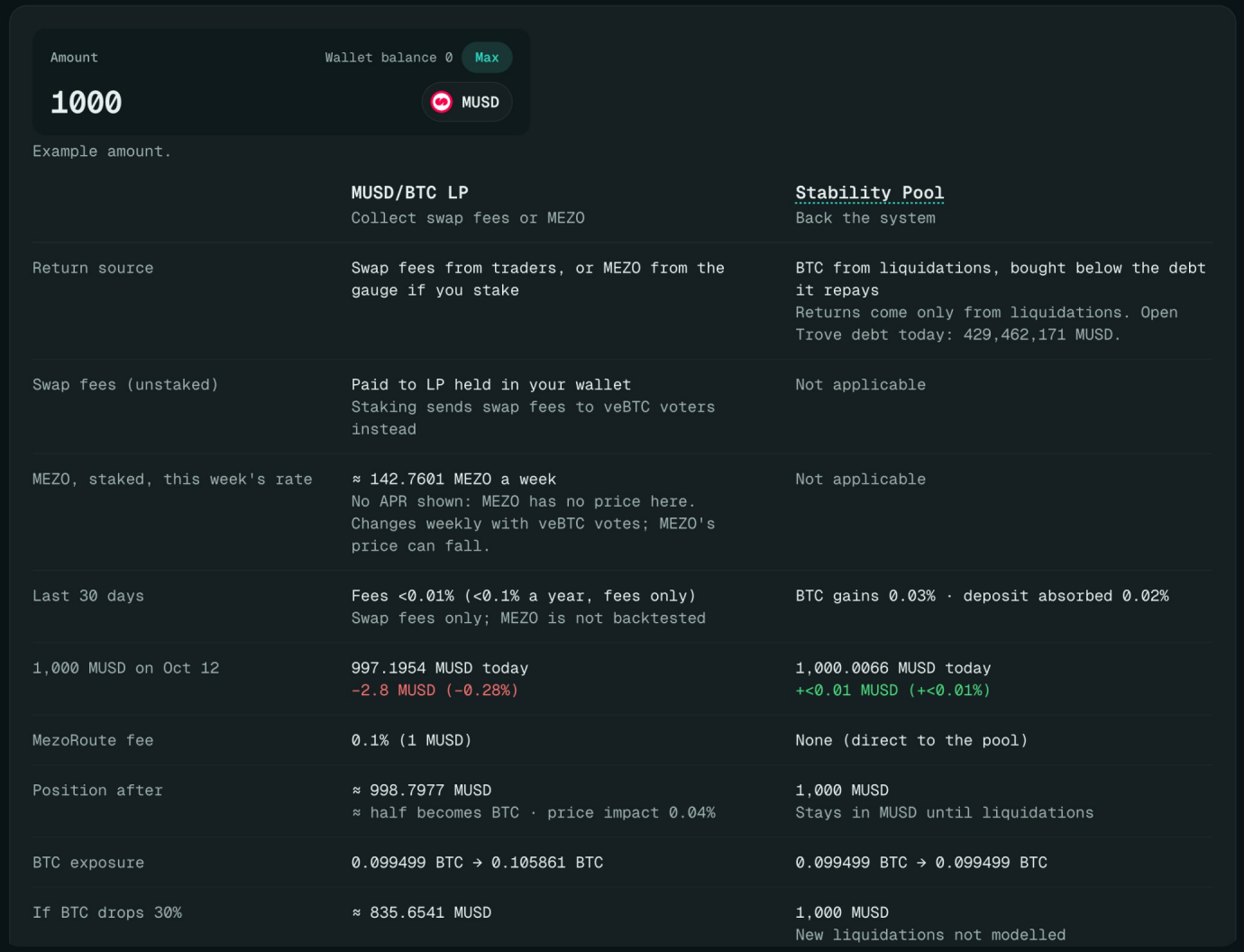
plus 2 gas-free signatures, on one screen. The swap size is computed exactly; leftovers return to the wallet.

Measured 19 Oct 2026 on testnet.mezo.org with 10 MUSD: 6 transactions sent, all succeeded. Exact approvals, so every new deposit repeats them. Sign-in excluded on both sides.

COMPARE ROUTES

Two routes, one BTC-risk view

- Return source: swap fees or MEZO vs liquidation BTC
- MEZO shown apart from fees, at this week's rate, never backtested
- Last 30 days from on-chain history, not a forecast
- BTC exposure after, and what a 30% BTC drop does

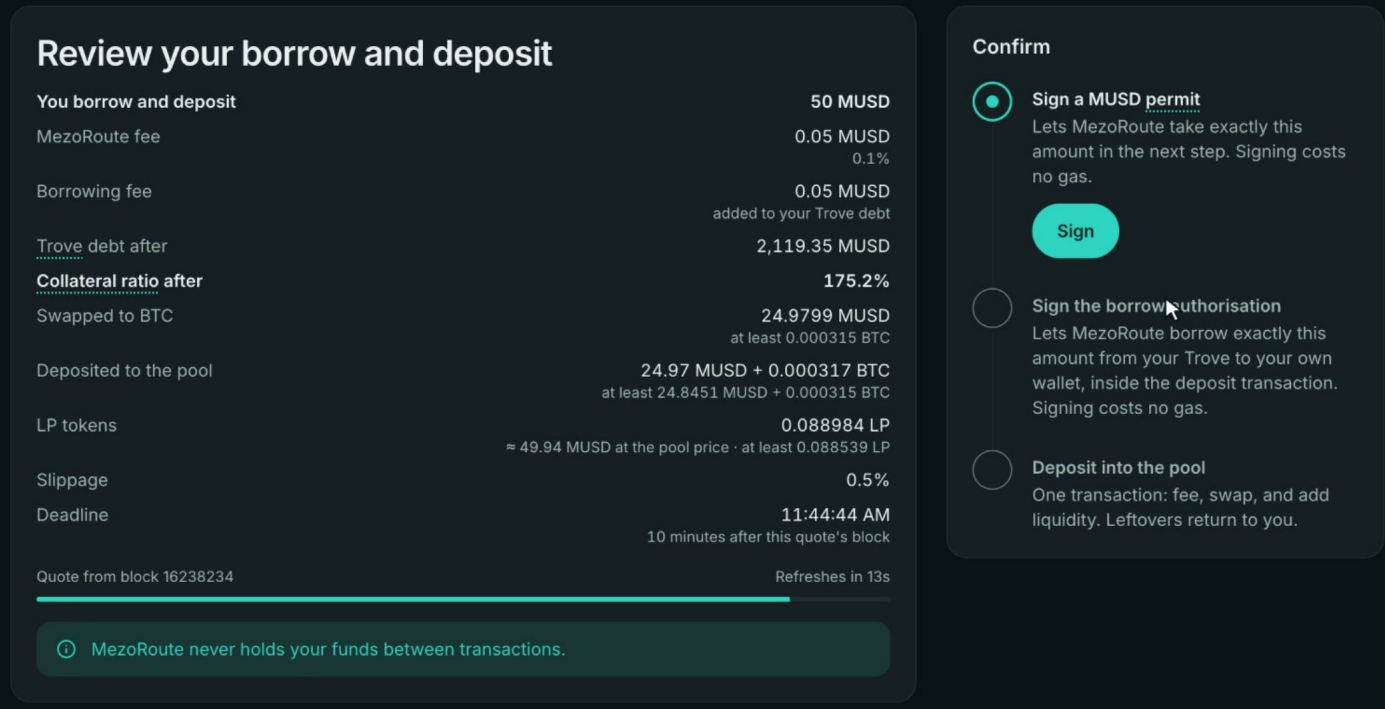


BORROW & DEPLOY

Borrow → LP in one transaction

1. Sign a MUSD permit
2. Sign the borrow authorisation (EIP-712, Mezo's BorrowerOperationsSignatures)
3. One transaction: borrow, swap the exact share, add liquidity, return leftovers

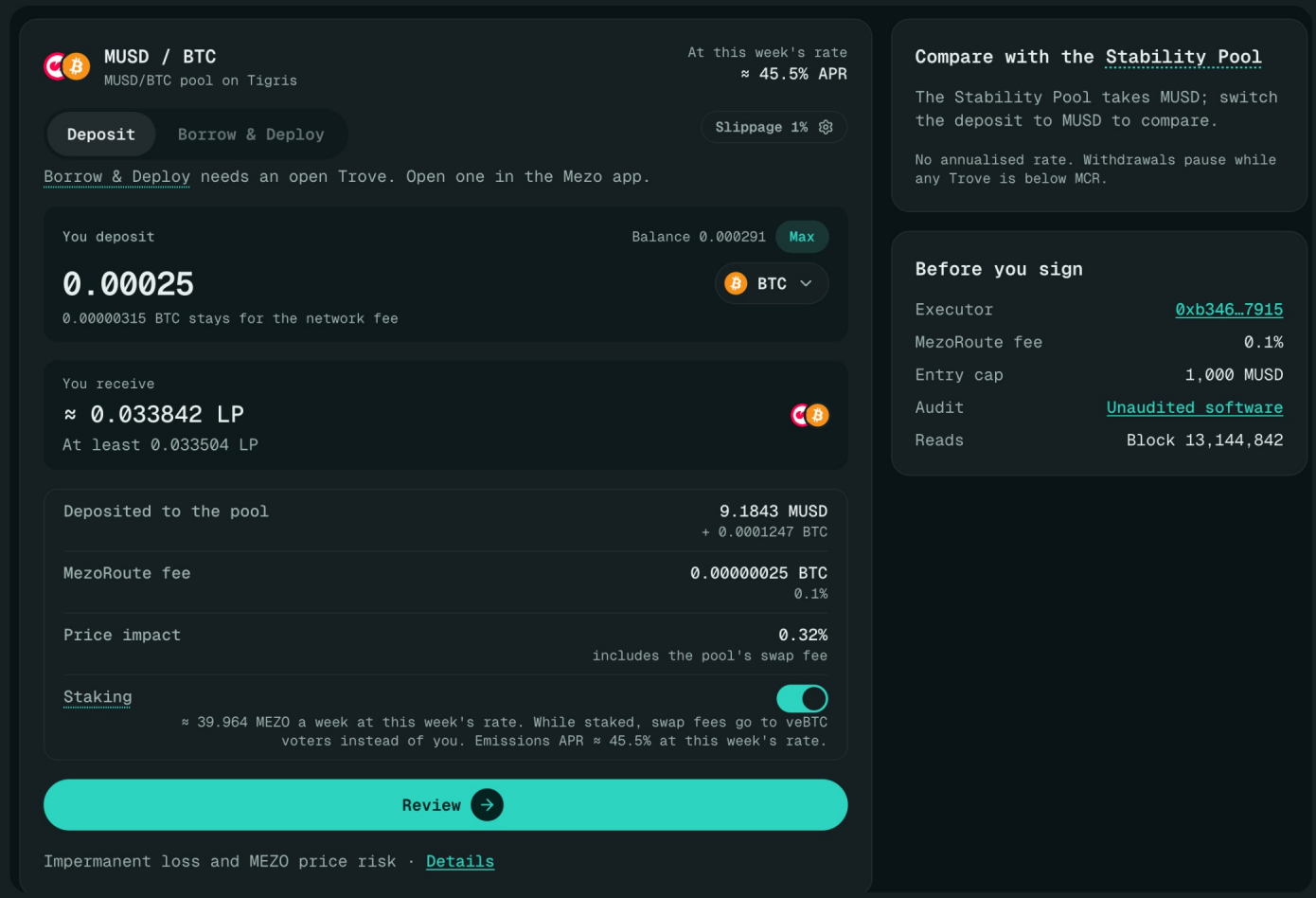
Minimum outputs, a deadline and a 160% collateral-ratio floor are checked before signing.



MEZO AND BTC ENTRY

Stake for MEZO, or start from BTC

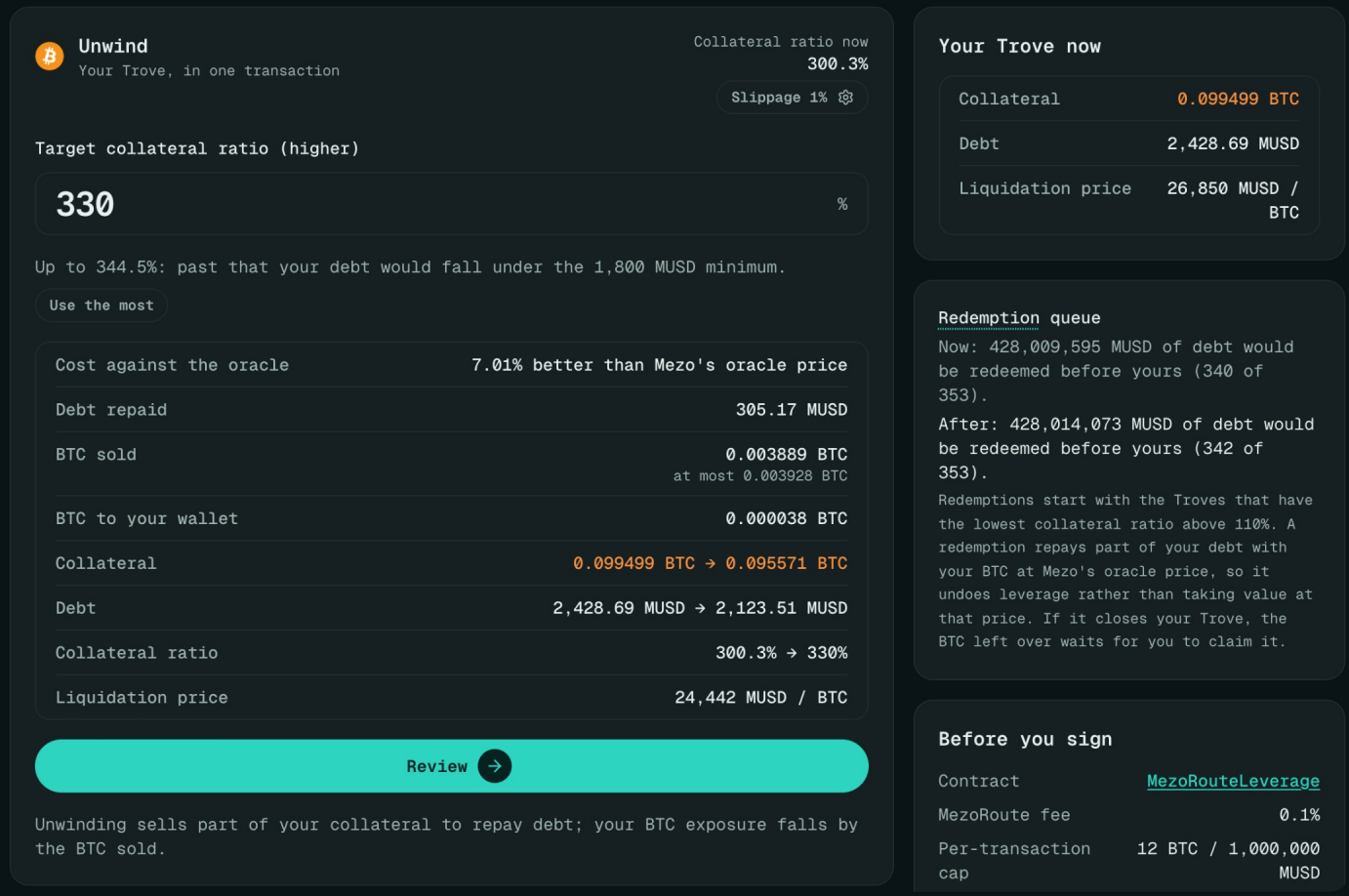
- Staking switch: the LP goes into the gauge in the same transaction
- Rate shown as this week's, never promised
- Staked LP gives its swap fees to veBTC voters; the app says so
- BTC in: about half becomes MUSD, gas reserve kept back



UNWIND · CLOSE · LEVERAGE

Your Trove, in one transaction

- **Unwind:** sell part of the collateral to repay debt
- **Close:** repay all debt from collateral, no MUSD needed
- **Leverage:** add BTC paid for by new debt
- Shown first: cost against Mezo's oracle and your place in the redemption queue



MezoRouteLeverage flash-swaps the MUSD/BTC pool and acts through Mezo's BorrowerOperationsSignatures with borrower = recipient = caller. Screenshot: testnet.

LEVERAGE, DONE CAREFULLY

What a Trove tool must handle

Risk	What MezoRoute does
Borrowing capacity	Plans a Refinance step first when only a refinance makes room
Recovery Mode	Blocks the flash actions with the reason; offers Repay from wallet
Pending redistribution	Reads getEntireDebtAndColl once; never adds it twice
Surplus after redemption	Claim surplus button (claimCollateral)
Sandwich risk	Every swap bounded by Mezo's PriceFeed: 5% default, 10% at most
Admin keys	None: immutable contract, no owner, no proxy
Funds at rest	Contract's MUSD and BTC balances unchanged after every call

Every suite runs under both pool token orders with the real Tigris pool · self-audit in contracts/SELF_AUDIT.md

MEZO INTEGRATION

Five Mezo surfaces, acted on, on chain

Surface	What MezoRoute does with it
Borrowing	Borrows, adjusts and refinances through BorrowerOperationsSignatures
Troves	Unwind, close, leverage; redemption queue by NICR; surplus claims
MUSD/BTC pool	Exact optimal swaps, flash swaps, LP entry and exit, fee claims
MEZO gauge	Stakes new LP in the same transaction; claim, stake, unstake
Stability Pool	Deposit, withdraw, claim BTC gains; Liquity maths for a 30-day backtest

> **MEZO, now integrated**
>
> In Wave 1 no gauge paid MEZO on this pool. It does now, so MezoRoute stakes for the user, and never holds the LP or the MEZO.

VALUE TO MEZO

Borrowers who stay, and a deeper pool

19.0M

MUSD supply on mainnet, down from 29.7M on 19 Oct

5

open Troves on mainnet; redemptions close them lowest ratio first

35.5k

MUSD in the basic MUSD/BTC pool, with 0.48 BTC

Unwind and Close give a Trove owner a way out before a redemption or a liquidation does it for them. Every LP entry deepens Mezo’s thinnest MUSD pool, now with MEZO on top.

Source: Mezo mainnet contract reads at block 13,144,716, 11 Nov 2026. The Stability Pool’s 14.17M MUSD is 99.9% protocol-owned.

TECHNICAL IMPLEMENTATION

Narrow contracts, tested to the wei

Contracts

- Three immutable contracts: no owner, admin or proxy; all verified
- Payouts are balance deltas inside the call; allowances reset to 0
- Per-transaction caps on mainnet; funds never rest in a contract
- 314 Foundry tests: fuzz, invariants, both pool token orders

App

- Quote maths mirrors each contract; mainnet entries replay exactly
- Results from events, never wallet balance changes
- Every error mapped to a message and a recovery action
- 1,663 unit tests; static Next.js export, no backend

Solidity 0.8.24 · OpenZeppelin 4.9 · Tigris · Next.js 14 · wagmi · viem · Mezo Passport · self-audit in contracts/SELF_AUDIT.md

PROJECT READINESS

Wave 2 round trips through the app

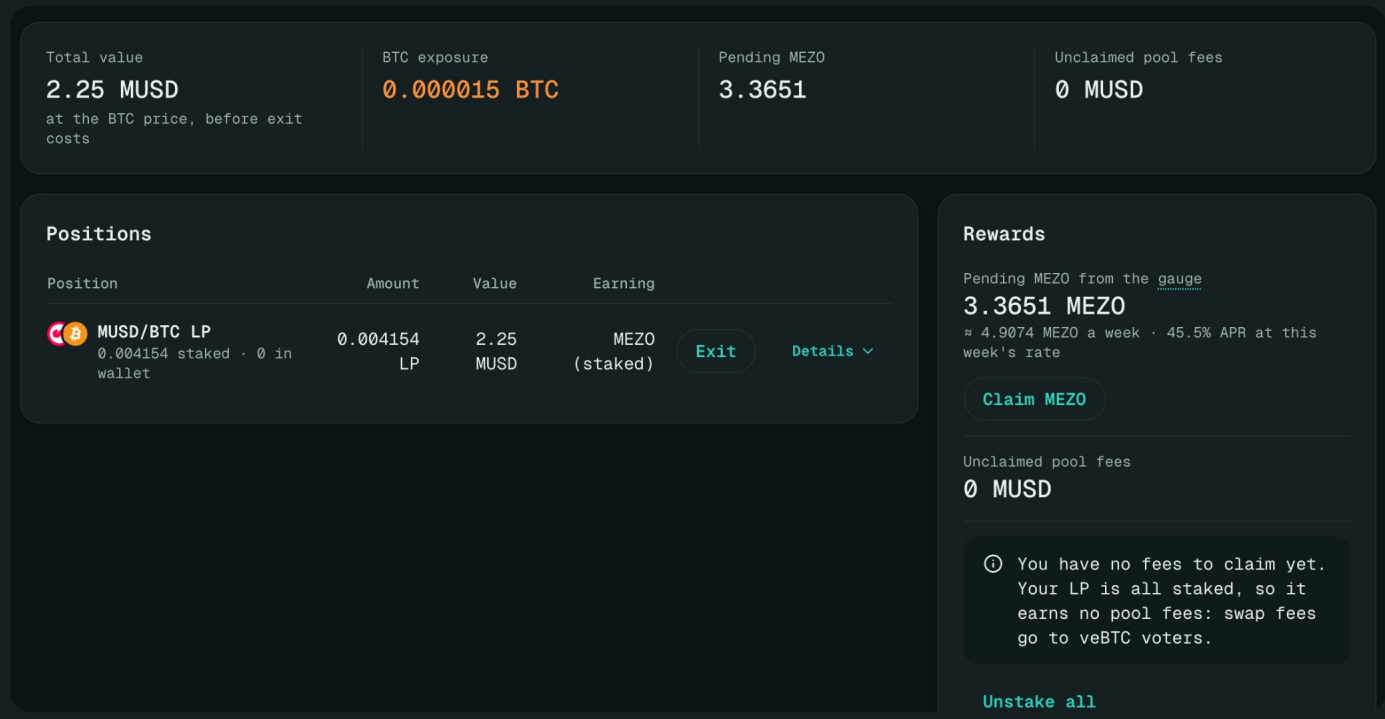
Network	Action	Result	Tx
Mainnet	LP entry + stake	10 MUSD → 0.017962 LP staked	0x415b…1445
Mainnet	Claim MEZO	0.001037 MEZO	0x093b…bd59
Mainnet	Unstake, then exit (Max)	9.9600 MUSD back	0x730c…89c2
Mainnet	Enter from BTC + stake	0.00003 BTC → 0.004154 LP	0xeac5…3caa
Testnet	Open with leverage	0.043 BTC → Trove at 201%	0x8eca…0f90
Testnet	Unwind	201% → 240%, 793.74 MUSD repaid	0x430f…b74c
Testnet	Leverage	240% → 201%, +0.009771 BTC	0x8d88…fff9
Testnet	Close	All 0.076518 BTC back, no MUSD used	0x381a…1b7b

Real MetaMask, 6–7 Nov 2026 · contracts held nothing afterwards · full hashes in PRODUCT_SPEC.md §20

USER EXPERIENCE

Bitcoin wallets, honest results

- Unisat, OKX and Xverse through Mezo Passport, plus EVM wallets
- Positions: wallet and staked LP in one row, pending MEZO, cost basis
- Plan an LP entry before connecting a wallet
- Light and dark; works on phones from 360 px



BUSINESS MODEL AND USERS

One disclosed fee, only when you add risk

0.1%

- On MUSD or BTC entering the LP route, and on Leverage
- Free: Unwind, Close, exits, Stability Pool
- Fixed at deploy, hard-capped at 0.5% in each contract
- Shown before signing and emitted on chain

Who it is for

- Mezo Trove owners who want to manage risk in one step
- BTC holders who want MEZO without buying MUSD first
- EVM or Bitcoin wallet, desktop or mobile
- Track 1 (DeFi) and Track 2 (access and distribution)

NEXT MILESTONES

Wave 3: outside users

Wave 3

Users beyond the team

Every mainnet transaction so far is our own wallet. A public stats page from contract events, and onboarding for Trove owners.

Wave 3

Mainnet Trove round trip

Unwind, Leverage and Close on a real mainnet Trove, and a Bitcoin wallet through Mezo's mainnet relayer.

After the hackathon

Audit, then scale

Static analysis and an external audit before raising caps; opening a Trove in the same flow.

> Shipped so far: both routes, Borrow & Deploy, MEZO staking, BTC entry, Unwind, Close and Leverage, live on mainnet.

Pick a route.
Know the BTC risk.

TRY IT

mezoroute.vercel.app

github.com/hms1499/MezoRoute

Unaudited software. Mainnet caps per transaction: 1,000 MUSD or 0.0125 BTC per entry; 0.0125 BTC or 2,500 MUSD per Trove action.